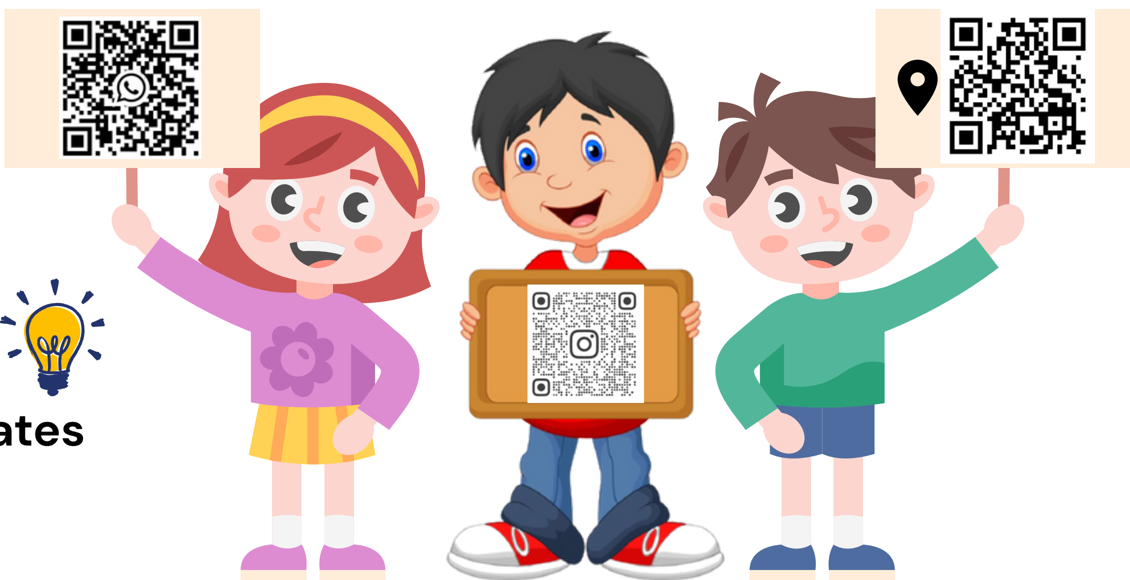
SATGURU STUDY CENTRE
Chennai - 81

TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025



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Class : 11

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UNIT TEST - 2, AUGUST- 2024

Time Allowed : 1.30 Hours | **PHYSICS** | Max. Marks : 35

PART - I

Answer all the questions 5x1=5

1. Force acting on the particle moving with constant speed is

   a) always zero b) need not be zero c) always non zero d) cannot be concluded

2. Two masses m_1 and m_2 are experiencing the same force where $m_1 < m_2$. The ratio of their acceleration a_1 / a_2 is

   a) 1 b) less than 1 c) greater than 1 d) zero

3. A Uniform force of $(2\hat{i} + \hat{j})$ N acts on a particle of mass 1Kg. The particle displaces from position $(3\hat{j} + \hat{k})$ m to $(5\hat{i} + 3\hat{j})$m. The workdone by the force on the particle is

   a) 9 J b) 6 J c) 10 J d) 12J

4. If the potential energy of the particle is $\alpha - \frac{\beta}{2}x^2$, then force experienced by the particle is

   a) $F = \frac{\beta}{2}x^2$ b) $F = \beta x$ c) $F = -\beta x$ d) $F = \frac{-\beta}{2}x^2$

5. The centre of mass of a system of particles does not depend upon,

   a) Position of Particles b) Relative distance between particles

   c) Massess of Particles d) Force acting on particle

PART - II

Answer any three questions. (Question Number 10 Is Compulsory) 3x2=6

6. Define One Newton.

7. State Lamis Theorem.

8. Why does a Cricketer pull his hands gradually in the direction of the ball's motion?

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9. What is Radius of Gyration?

10. Calculate the energy consumed in electrical units when a 75W fan is used for 8 hours daily for 1 month? (30 days).

PART - III

III Answer any three of the following. Q.No.15 is Compulsory. **3x3=9**

11. Which is easier to move on object by Push or Pull? Explain with the help of Free body diagram.

12. State Newton's three Laws of Motion.

13. Distinguish between Elastic Collision and Inelastic Collision. (any three points)

14. Derive the relation between Torque and Angular Acceleration.

15. An object of mass 10 Kg moving with a speed of $15ms^{-1}$ hits the wall an comes to rest within a) 0.03 S b) 10 S.Calculate the Impulse and Average force acting on the object in both the cases.

PART - IV

IV Answer the following in detail: **3x5=15**

16. a) Explain the motion of blocks connected by a string in Vertical Motion.

(OR)

b) Show that in an Inclined plane, Angle of Friction is equal to Angle of repose.

17. a) State and Explain Work Energy principle. Mention any three examples for it.

(OR)

b) Arrive at an expression for Elastic Collision in one dimension. Find the final Velocities v_1 and v_2

18. a) State and Prove Parallel axis Theorem.

(OR)

b) Derive the expression for moment of Inertia of a uniform ring about an axis passing through the Centre and Perpendicular to the plane.